

1. **Explain the basic architecture of the Internet. Describe the roles of end systems, access networks, routers, and communication links in providing Internet connectivity.**
2. **Explain the concept of network edge and network core. Discuss the different types of access networks used to connect end systems to the Internet.**
3. **Explain the architectural layers of the TCP/IP model with a neat block diagram, clearly detailing the functions of each layer during encapsulation and decapsulation.**
4. **What is a protocol? Explain the key elements of a network protocol and discuss why protocols are necessary for communication between networked systems.**
5. **Describe the structure and purpose of a Uniform Resource Identifier (URL) by identifying its four core components.**
6. **Explain the different layers of the Internet protocol stack. Describe the functions and responsibilities of each layer with suitable examples of protocols.**
7. **Explain the key differences between non-persistent and persistent HTTP connections. How do persistent connections improve web communication efficiency?**
8. **Explain the Domain Name System (DNS). Describe the need for DNS, hierarchical naming structure, DNS servers, and the process of resolving a domain name into an IP address.**
9. **Explain the concept of protocol layering and list the two fundamental principles governing protocol layering.**
10. **Differentiate between the Client-Server paradigm and the Peer-to-Peer (P2P) paradigm.**
11. **Explain the World Wide Web and HTTP. Describe the structure of an HTTP request and HTTP response message with suitable examples.**
12. **Explain FTP (File Transfer Protocol). Describe its operation, control connection, data connection, and the process of uploading and downloading files.**
13. **Explain the complete scenario of an electronic mail system transferring a message from a sender to a receiver.**
14. **Explain the Client-Server paradigm in detail. Describe the roles of the client and server, and explain how a client communicates with a server using sockets.**
15. **Explain the Peer-to-Peer (P2P) paradigm with the help of BitTorrent as a case study. Describe the roles of peers, trackers, chunks, and the process of downloading and uploading files in a BitTorrent network.**